1 Integration by Parts

1. Consider the integral $\int x\sin(x)dx$.

   (a) Explain why substitution cannot be used to evaluate this integral.

   (b) Recall the Product Rule of differentiation says that $\frac{d}{dx}(fg)(x) = f'(x)g(x) + f(x)g'(x)$. Let $f(x) = x$ and $g'(x) = \sin(x)$. Use the Product Rule to rewrite the integrand, then evaluate the integral.

   (c) Repeat what you did in part (b) for the integral $\int f(x)g'(x)dx$ and simplify as much as possible. The technique of rewriting the integral in this way is called *Integration by Parts*.

2. In part (c) of Question 1, you proved the *Integration by Parts* formula

$$\boxed{\int f(x)g'(x)dx = f(x)g(x) - \int f'(x)g(x)dx}.$$

The following notation is also common. Let $u = f(x)$ and $v = g(x)$. Then $\int f(x)g'(x)dx = \int udv$. Use the Substitution Rule to prove that, in this notation, the formula for Integration by Parts becomes

$$\boxed{\int udv = uv - \int vdu}.$$

Strategy. When deciding on a choice for u and dv, try to choose $u = f(x)$ to be a function that becomes simpler when differentiated (or at least not more complicated) and $dv = g'(x)dx$ to be a function which is readily integrated to give v.

3. Evaluate $\int xe^x dx$.

4. Evaluate $\int \ln(x)dx$.

5. Prove that if f' and g' are continuous, then

$$\boxed{\int_a^b f(x)g'(x)dx = f(b)g(b) - f(a)g(a) - \int_a^b g(x)f'(x)dx}.$$

6. Calculate $\int_0^1 \tan^{-1}(x)dx$.

2 Repeated Integration by Parts

7. Evaluate $\int e^x \sin(x)dx$ by integrating by parts twice.

8. Evaluate $\int x^2 e^x dx$ by integrating by parts twice.

3 Tabular Integration

For an integral $\int udv = \int f(x)g'(x)dx$ where repeated differentiation of u eventually yields zero, repeated Integration by Parts is most easily implemented via a table. This method is called *Tabular Integration*.

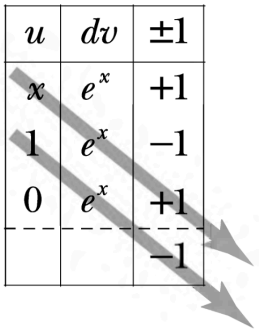
Begin by making a table with three columns:

- The first column consists of $u = f(x)$ and its subsequent derivatives.
- The second column consists of $g'(x)$ and its subsequent antiderivatives.
- The final column alternates between $+1$ and -1, beginning with $+1$.

Example 3.1. Consider $\int xe^x dx$ from Question 3. Repeatedly differentiate $f(x) = x$ and repeatedly antidifferentiate $g(x) = e^x$ to obtain the table below. Note that the rightmost column has one more row than the other two columns.

u	dv	± 1
x	e^x	$+1$
1	e^x	-1
0	e^x	$+1$
		-1

Multiply each term in the left column with the other values along a downward diagonal, as illustrated below:



Adding "$+C$", this gives $\int xe^x dx = xe^x - e^x + C$, which should agree with your answer to Question 2.

9. Use Tabular Integration to evaluate $\int x^2 e^x dx$. Verify that your answer agrees with your answer to Question 8.

10. Use Tabular Integration to evaluate $\int x^2 \sin(x) dx$.

4 Proof of the Tabular Integration Formula

Given two functions f, g defined on an interval I, let $f = f^{(0)}, f^{(1)}, f^{(2)}, \ldots, f^{(n)}$ denote the first n derivatives of f and $g = g^{(0)}, g^{(-1)}, g^{(-2)}, \ldots, g^{(-n)}$ denote the first n antiderivatives of g.

Theorem 4.1 (Generalized Integration by Parts Formula)**.** *Let f be of class C^n on an interval I (meaning that $f^{(n)}$ exists and is continuous on I) and let g be of class C^1 on I. Then, for $n \in \mathbb{N}$,*

$$\int f(x)g^{(1)}(x)dx = \sum_{j=0}^{n-1}(-1)^j f^{(j)}(x)g^{(-j)}(x) + (-1)^n \int f^{(n)}(x)g^{(-n+1)}(x)dx. \tag{4.2}$$

We now prove Theorem 4.1 by induction on n. [1]

11. Show that, for $n = 1$, Equation (4.2) reduces to the basic Integration by Parts formula. (This establishes the base of the induction.)

12. Now suppose that Equation (4.2) holds for some $n \geq 1$ (this is the inductive hypothesis). Prove that this implies that Equation (4.2) holds for $n+1$, completing the proof.

[1] Consider a subset S of $\mathbb{N}$. Note that if S has the property that (1) $1 \in S$ and (2) $n \geq 1 \in S$ implies $n+1 \in S$, then $S = \mathbb{N}$ (since $1 \in S$ and $1 \in S$ implies $2 \in S$ which implies $3 \in S$ which implies $4 \in S$, and so on). Consider now the set $S = \{n \in \mathbb{N} : P(n) \text{ is true}\}$. If we can show that (1) $P(1)$ is true (i.e., $1 \in S$) and (2) $P(n)$ is true implies $P(n+1)$ is true ($n \in S$ implies $n+1 \in S$), then we will have proved that $P(n)$ is true for all n (i.e., $S = \mathbb{N}$). This technique is called *Proof by Induction*.

Corollary 4.3. *If* $f^{(n)} = 0$, *then* $\int f(x)g^{(1)}(x)dx = \sum_{j=0}^{n-1}(-1)^j f^{(j)}(x)g^{-j}(x) + C$.

13. Prove Corollary 4.3.